SRS

Software Requirements Specification (SRS)

> **Project:** Endur (Performance Review & Feedback Management System)

> **Team:** 39_endur

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1. Preface

1.1 Expected Readership

This document is intended for:

- **Project Developers (Team 39_endur):** To understand the technical specifications and constraints.
- **Domain Expert (Dean of Academics):** To validate that the operational workflows match university policy.
- **Project Evaluators:** To assess the depth of requirement analysis and system modeling.

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2. Introduction

2.1 Need for the System

The current end-semester feedback model fails on two critical fronts:

> **Pain Point 1: Timing Latency**

> Feedback arrives too late to benefit the current batch and is often skewed by "Revenge Ratings" submitted after grades are released.

> **Pain Point 2: Engagement Fatigue**

> The Dean reports that **only ~30% of students participate** because the process is viewed as a "chore" rather than a meaningful contribution.

2.2 System Functions

- **Continuous Feedback Cycles:** Allows multiple review windows per semester.
- **Role-Based Dashboards:** Distinct views for Student (Submission), Faculty (Reflection), and HOD (Oversight).

- **Gap Analysis:** Automated comparison between Faculty Self-Reflection and Student Perception.
- **Anonymity Preservation:** Decoupling of student identity from feedback data.

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3. Glossary

Refer to definitions.yaml in the repository for the complete Ubiquitous Language.

- **FeedbackCycle:** A defined, time-bound period for data collection.
- **GapAnalysis:** The quantitative difference between a professor's self-rating and the students' average rating.
- **AnonymizedReport:** A generated view where student identities are stripped to ensure psychological safety.
- **ActionReport:** A formal plan submitted by faculty to the HOD to address performance gaps.
- **ReviewOfReviews:** A meta-evaluation process accessible to both students and faculty members that assesses the effectiveness, fairness, and usefulness of the feedback system itself, including question design and timing.
- **FeedbackTrend:** A longitudinal view of PerformanceScores across multiple FeedbackCycles, used to evaluate improvement, stagnation, or decline over time.
- **SelfReflection:** A reflective input provided to a FacultyMember and Student to assess their own performance, challenges, and improvements during a semester.
- **ReviewCheckIn:** A scheduled, lightweight review interaction between a FacultyMember and the DepartmentHead (or academic reviewer) focused on discussing feedback trends, gap analysis, and progress on previously committed action items, rather than re-evaluating raw scores.
- **FeedbackResponse:** A single, immutable submission made by a Student during a FeedbackCycle for a specific CourseOffering, consisting of PerformanceScores mapped to EvaluationParameters. Each FeedbackResponse represents one complete feedback instance and is used as an atomic unit for aggregation, analysis, and auditing.

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4. User Requirements Definition

High-level services provided for the user, described in natural language.

4.1 Student Services

- **UR-01:** Students shall be able to submit weekly/monthly FeedbackCycle which are lightweight (taking < 30 seconds) to ensure high participation.

- **UR-02:** Students shall be able to view a history of their past submissions for transparency.
- **UR-03:** Students should be able to fill monthly feedback form on ReviewOfReviews.

4.2 Faculty Services

- **UR-04:** Faculty shall be able to submit a **SelfReflection** assessment before viewing student feedback.
- **UR-05:** Faculty shall be able to view their performance and FeedbackTrends.
- **UR-06:** Faculty shall be able to submit an **ActionReport** if their performance scores trigger a system alert.

4.3 Administrative (HOD) Services

- **UR-07:** The Department Head (HOD) shall be able to view aggregated reports for all faculty members in their departments.
- **UR-08:** The HOD shall be able to conduct ReviewCheckIn with the concerned faculty members.
- **UR-09:** The HOD is able to define **EvaluationParameter** for every FeedbackCycle.

4.4 Dean Services

- **UR-10:** The Dean shall be able to view institution-wide performance reports covering all departments, faculty members, and courses.
- **UR-11:** The Dean shall be able to act as the approving authority when a conflict of interest is identified, such as when a Department Head is reviewing their own performance.
- **UR-12:** The Dean shall be able to view ReviewCheckIn records and ActionReports from any department for oversight purposes.
- **UR-13:** The Dean shall be able to track long-term FeedbackTrends across academic years to identify overall strengths, risks, and areas needing improvement at the institutional level.
- **UR-14:** The Dean shall be able to review ComplianceAudit logs and flagged feedback submissions to ensure fairness, integrity, and adherence to academic policies.
- **UR-15:** The Dean shall be able to freeze, reopen, or invalidate FeedbackCycles in exceptional academic or administrative situations, with all such actions recorded in the audit log.

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5. System Requirements Specification

5.1 Functional Requirements (FR)

FR-01: FeedbackCycle Window Enforcement

The system shall enforce FeedbackCycle submission boundaries based on configured start and end timestamps.

- Feedback submission requests received before `FeedbackCycle.startTimestamp` or after `FeedbackCycle.endTimestamp` shall be rejected.
- Rejected submissions shall return a validation error indicating that the FeedbackCycle is inactive.
- Enforcement shall apply uniformly across all FeedbackCycle types.

FR-02: Attendance-Weighted PerformanceScore Calculation

The system shall calculate PerformanceScore using AttendanceWeightedFeedback when attendance data is available for a CourseOffering.

- If student attendance exceeds 75%, the applied weight shall be 1.0.
- Attendance thresholds and corresponding weights shall be configurable.
- If attendance data is unavailable, unweighted averaging shall be applied.
- Weighting logic shall be applied per EvaluationParameter.

FR-03: GapAnalysis Computation

The system shall compute the gap using the following logic:

$$Gap = |SelfReflection.Score - \text{Average}(StudentPerformanceScore)|$$

FR-04: ActionReport Immutability

The system shall enforce immutability of ActionReports after submission.

- Once submitted, an ActionReport shall become read-only to the FacultyMember.
- Editing, deletion, or overwriting of a submitted ActionReport shall not be permitted.
- A new ActionReport may only be submitted in a subsequent FeedbackCycle.
- DepartmentHead access to ActionReports shall be read-only.

FR-05: AnonymizedReport Generation

The system shall generate AnonymizedReports for each FacultyMember and CourseOffering after the closure of a FeedbackCycle.

- AnonymizedReports shall aggregate PerformanceScores per EvaluationParameter.
- Individual student identities and raw submissions shall not be exposed.
- QualitativeFeedback shall be grouped and anonymized before inclusion.
- AnonymizedReports shall be immutable once generated.

FR-06: FeedbackTrend Computation

The system shall compute FeedbackTrends across multiple FeedbackCycles for the same CourseOffering and FacultyMember.

- FeedbackTrends shall be based on aggregated PerformanceScores.
- Trends shall be calculated per EvaluationParameter.
- Historical FeedbackTrends shall not be recalculated when new cycles are added.
- FeedbackTrends shall not alter historical data.

FR-07: ComplianceAudit for Feedback Integrity

The system shall perform ComplianceAudit checks on feedback submissions to detect integrity violations.

- The system shall flag **RandomTicking** patterns, including:

  - Identical PerformanceScores across all EvaluationParameters.

  - Multiple submissions within an unusually short time window.

- Flagged submissions shall be excluded from PerformanceScore calculations.
- ComplianceAudit results shall be logged for administrative review.

FR-08: ReviewCheckIn Record Management

The system shall store ReviewCheckIn records associated with a FacultyMember and CourseOffering.

- ReviewCheckIn records shall reference:

  - Relevant FeedbackTrends

  - Associated GapAnalysis results

  - Related ActionReports

- ReviewCheckIn records shall be immutable once saved.

FR-09: Institutional Performance Aggregation

The system shall generate institution-wide analytics for the Dean by aggregating data across all Departments.

- The system shall compute an overall **PerformanceScore** for the entire institution by averaging aggregated scores from all AnonymizedReports.
- The system shall allow the Dean to filter **FeedbackTrends** by:

  - Department

  - Academic Year

  - CourseContentQuality

- The system shall generate a comparative view identifying the **highest and lowest performing Departments** based on aggregated EvaluationParameters.

FR-10: Administrative Conflict Resolution Workflow

The system shall enforce a specific workflow when a DepartmentHead is the subject of a review.

- If the FacultyMember associated with a CourseOffering is also the current DepartmentHead, the system shall automatically reassign **ReviewCheckIn approval authority** to the Dean.
- The system shall prevent the DepartmentHead from viewing or interacting with their own **ActionReport** as a reviewer.

FR-11: Emergency Cycle Management

The system shall allow the Dean to manually override the state of a FeedbackCycle in exceptional circumstances.

- The Dean shall be able to **Freeze** an active FeedbackCycle, immediately rejecting new FeedbackResponse submissions regardless of the configured end timestamp.
- Any manual state change triggered by the Dean shall automatically generate a **ComplianceAudit log entry** requiring a justification note.

FR-12: Long-Term Trend Analysis

The system shall compute multi-year FeedbackTrends to support institutional strategic assessment by the Dean.

- The system shall aggregate **PerformanceScores** across multiple academic years to visualize long-term progression.
- The system shall highlight trends in **CourseContentQuality** independently from delivery-related EvaluationParameters.
- Historical trend data shall remain immutable once calculated.

5.2 Non-Functional Requirements (NFR)

- **NFR-01 (Anonymity & Privacy):**

The system shall ensure that Student_ID is never retrievable, inferable, or reconstructible from any FeedbackResponse, AnonymizedReport, or downstream analytics store.

- **NFR-02 (Scalability):**

The system shall support a minimum of *500 concurrent write operations* within the final *10 minutes of an active FeedbackCycle* without data loss or degradation of response correctness.

- **NFR-03 (Availability):**

The system shall maintain *99.9% service availability* during institution-defined peak periods, including active exam weeks and FeedbackCycle submission windows.

- **NFR-04 (Immutability & Data Integrity):**

The system shall enforce immutability for all finalized artifacts, including submitted feedback responses, generated AnonymizedReports, ActionReports, and ReviewCheckIn records.

- **NFR-05 (Auditability):**

The system shall maintain an append-only audit log capturing all create and state-transition events.

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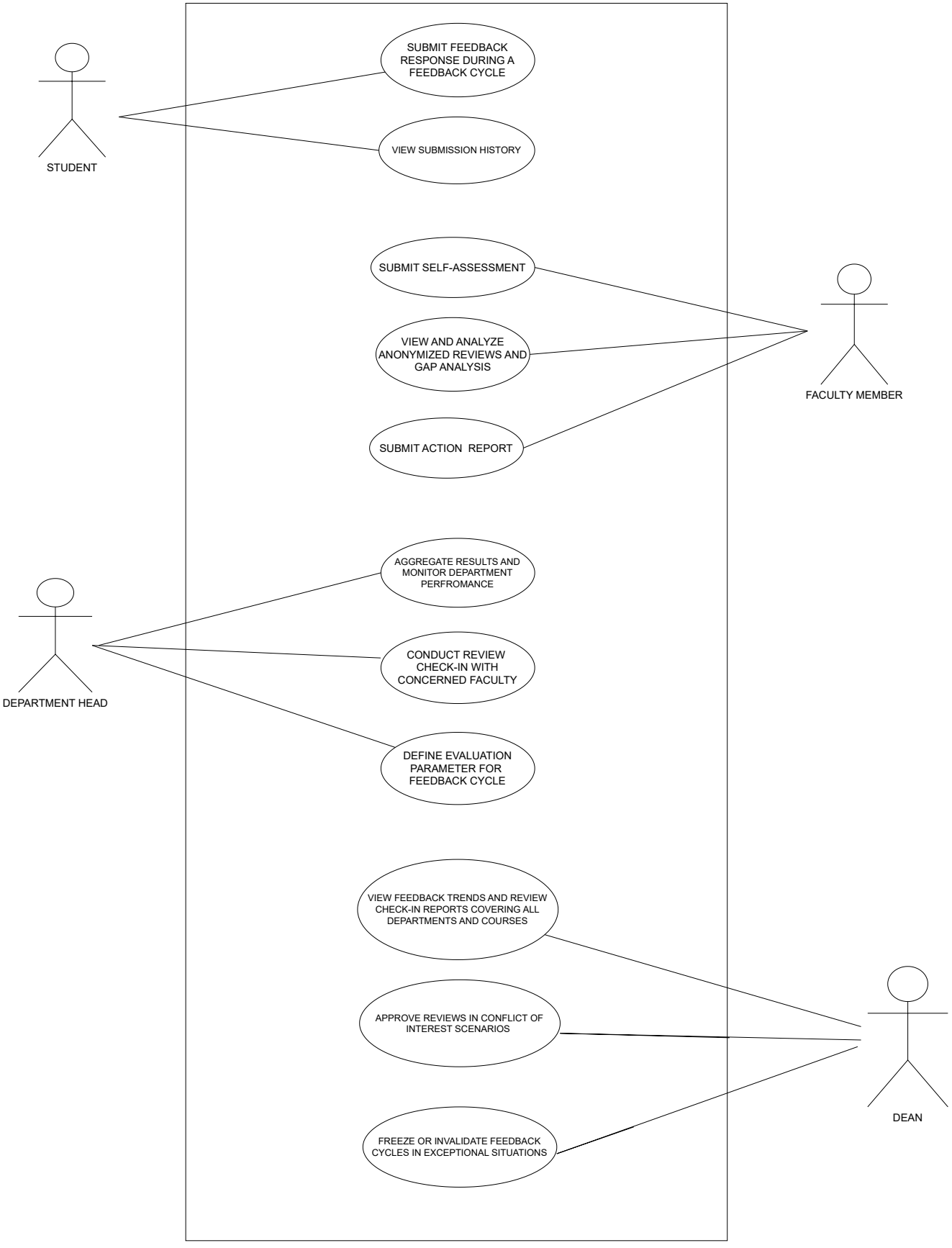
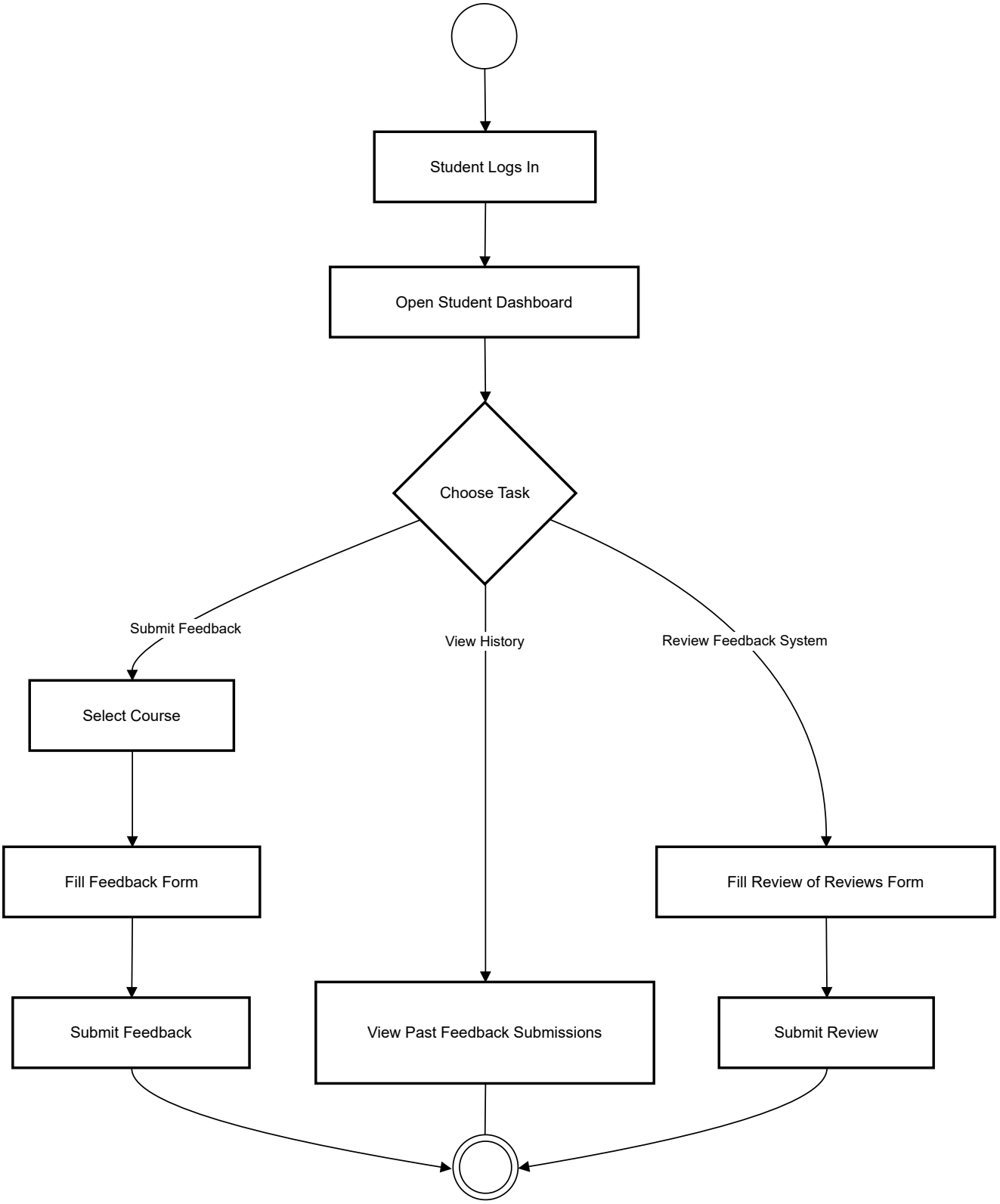
6. System Models

6.1 Use Case Diagram

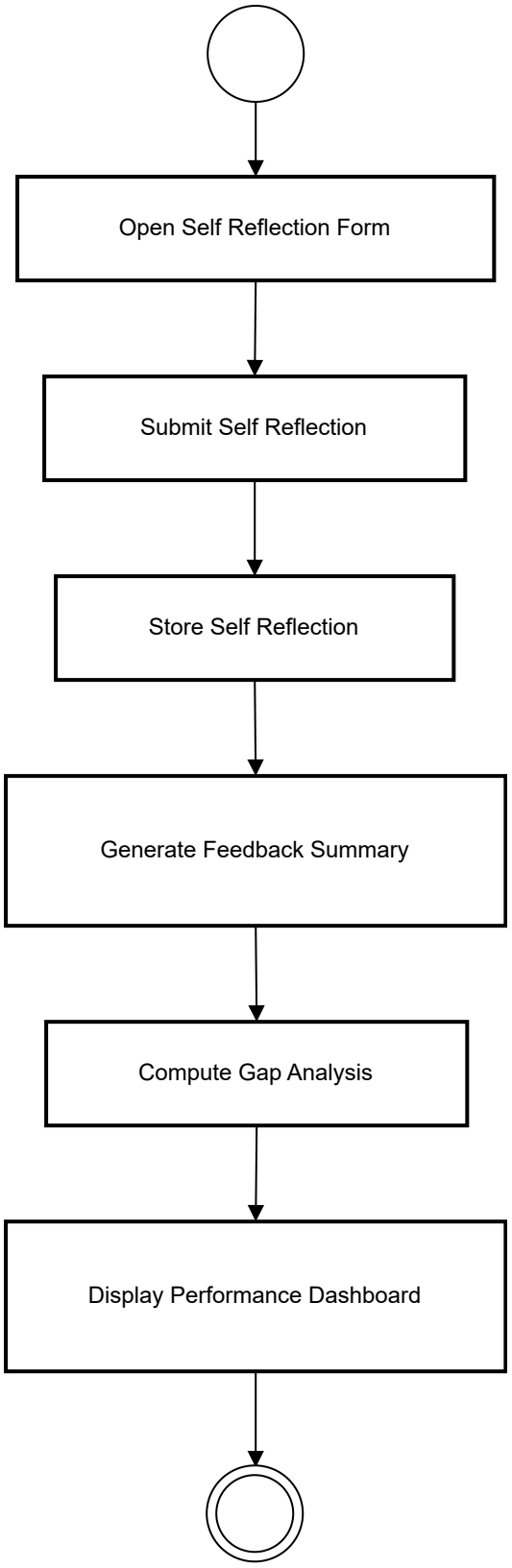


6.2 Activity Diagrams

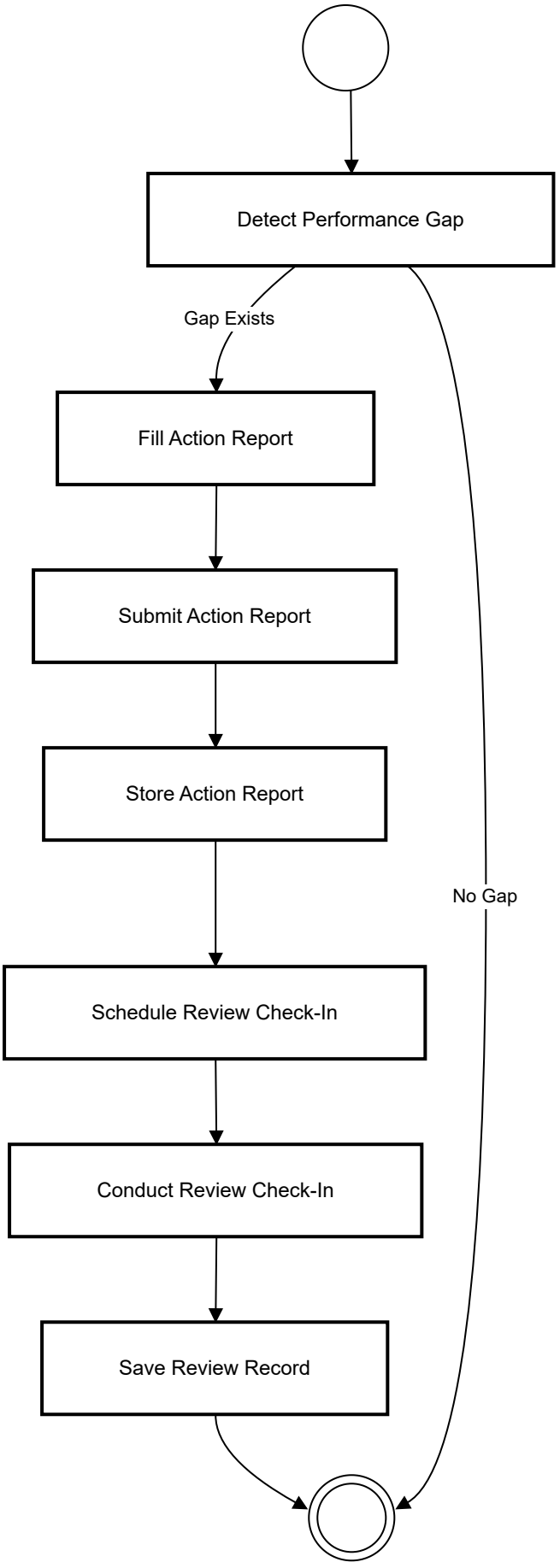
1) Student Feedback Submission



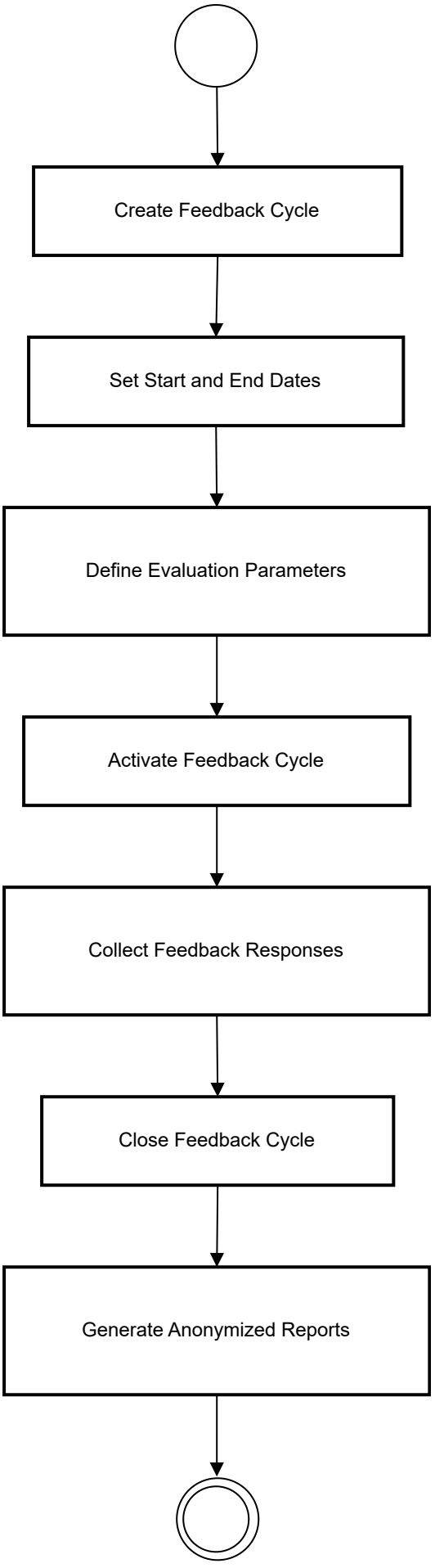
2) Faculty Self-reflection and Gap-Analysis



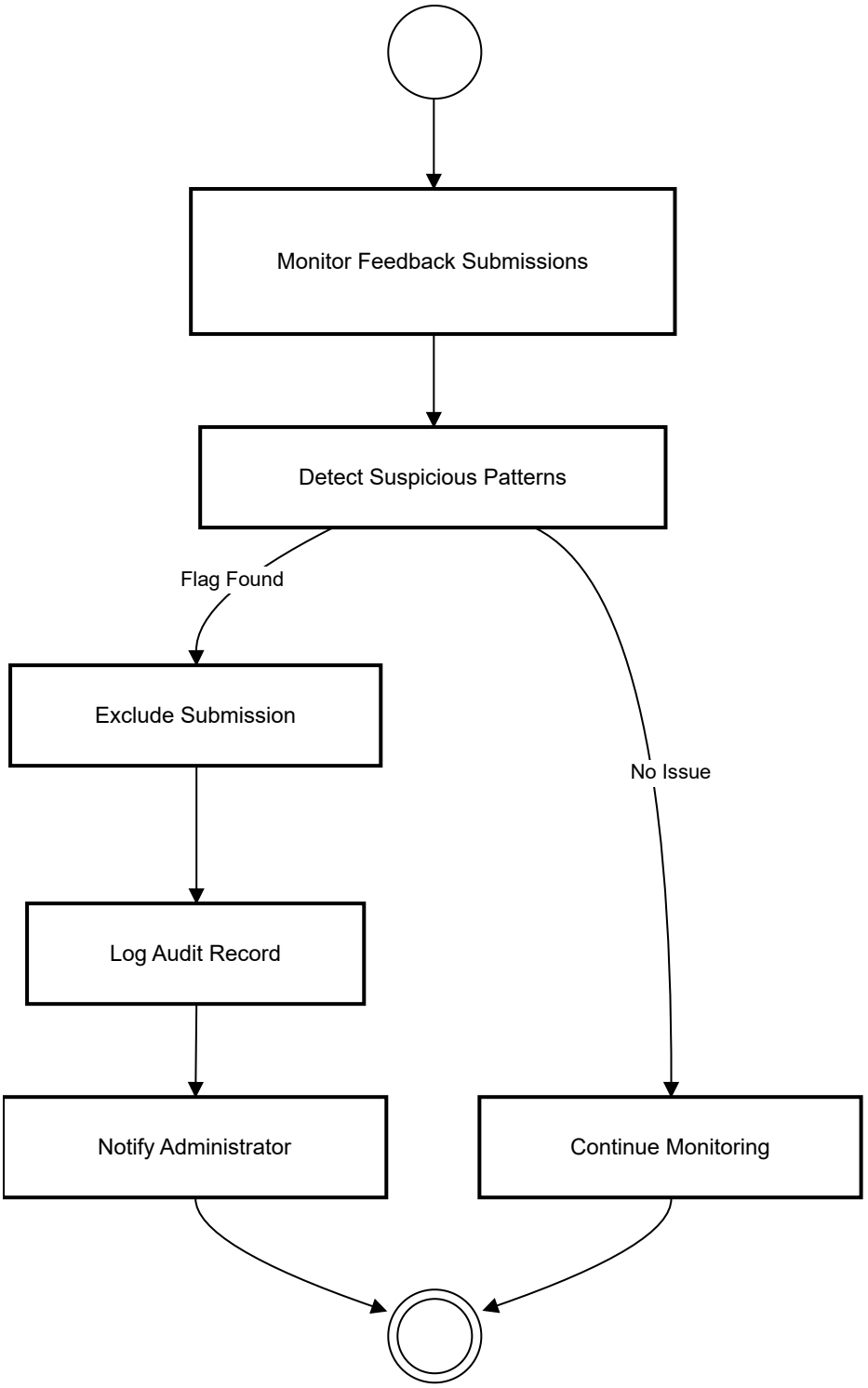
3) Action Report and Review Check In



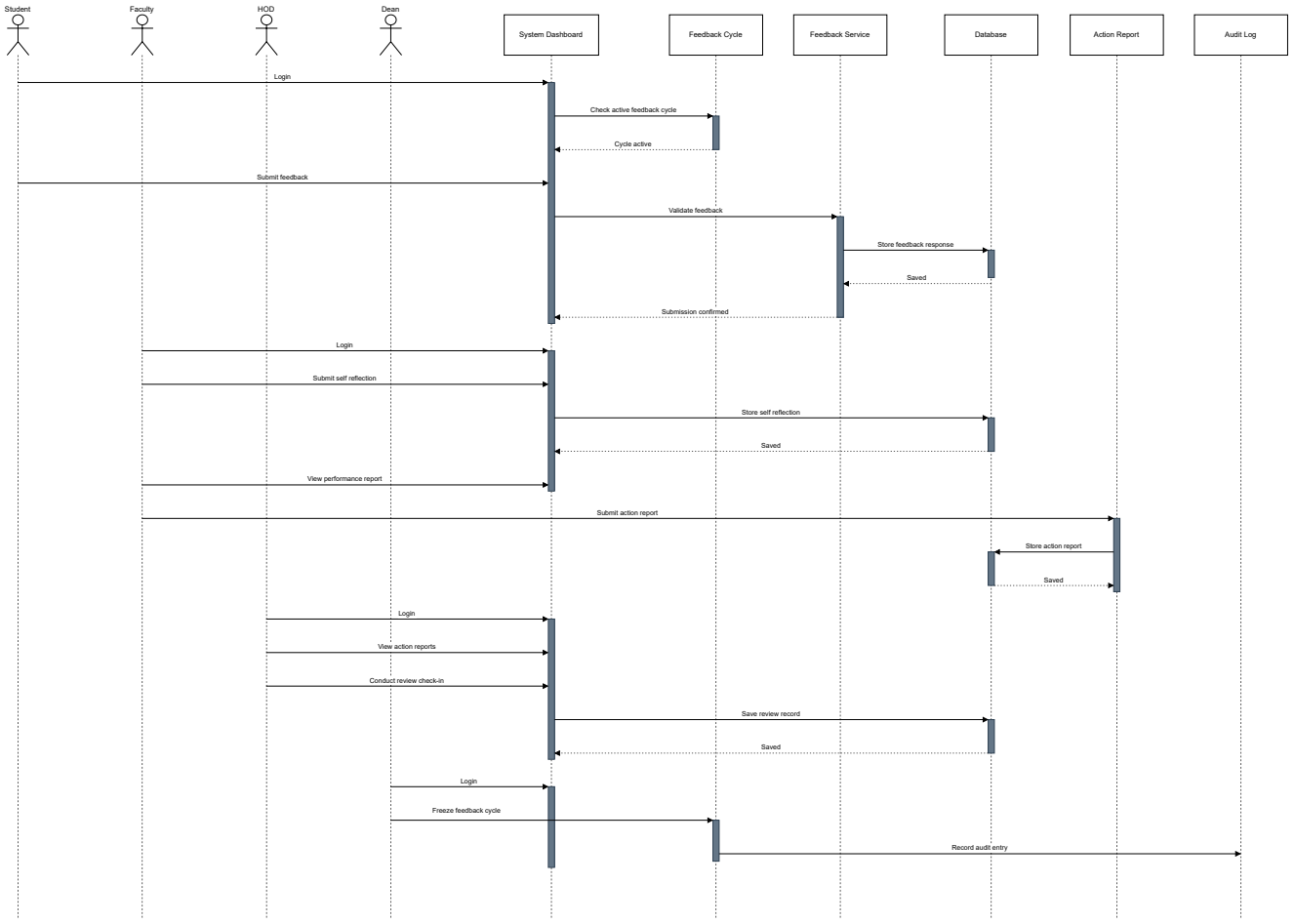
4) Feedback Cycle Management



5) Compliance Audit



6.3 Sequence Diagram



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